

MatForge

Enhanced Material System for
Vulkan Ray Tracer

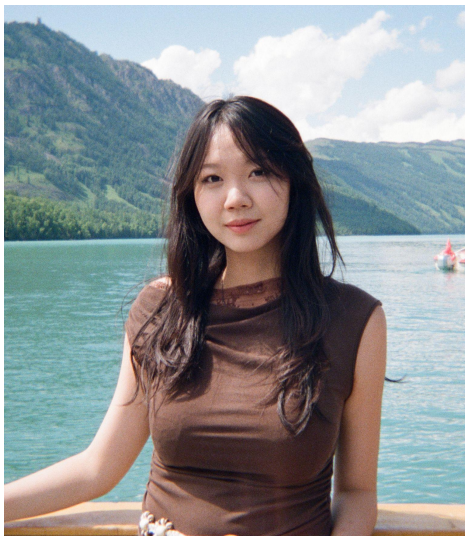
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About Us



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The Problem

- **Sampling:** Random clustering → slow convergence
- **Geometry:** Tessellation → 3× memory, 11× slower
- **Materials:** 15-40% wasted samples (rejections)
- **Energy:** Rough surfaces too dark (100× error)

Why it matters

- **First** combined 4-paper implementation
- Production-quality Vulkan path tracer
- Real-time compatible



Full Rendering Pipeline Enhanced

1. Quad-Optimized LDS
 2. RMIP
 3. Bounded VNDF
 4. Fast-MSX
- Sampling Foundation
 - Geometry Detail
 - Direction Sampling
 - BRDF Evaluation

Quad-Optimized LDS

Generate low-discrepancy random numbers

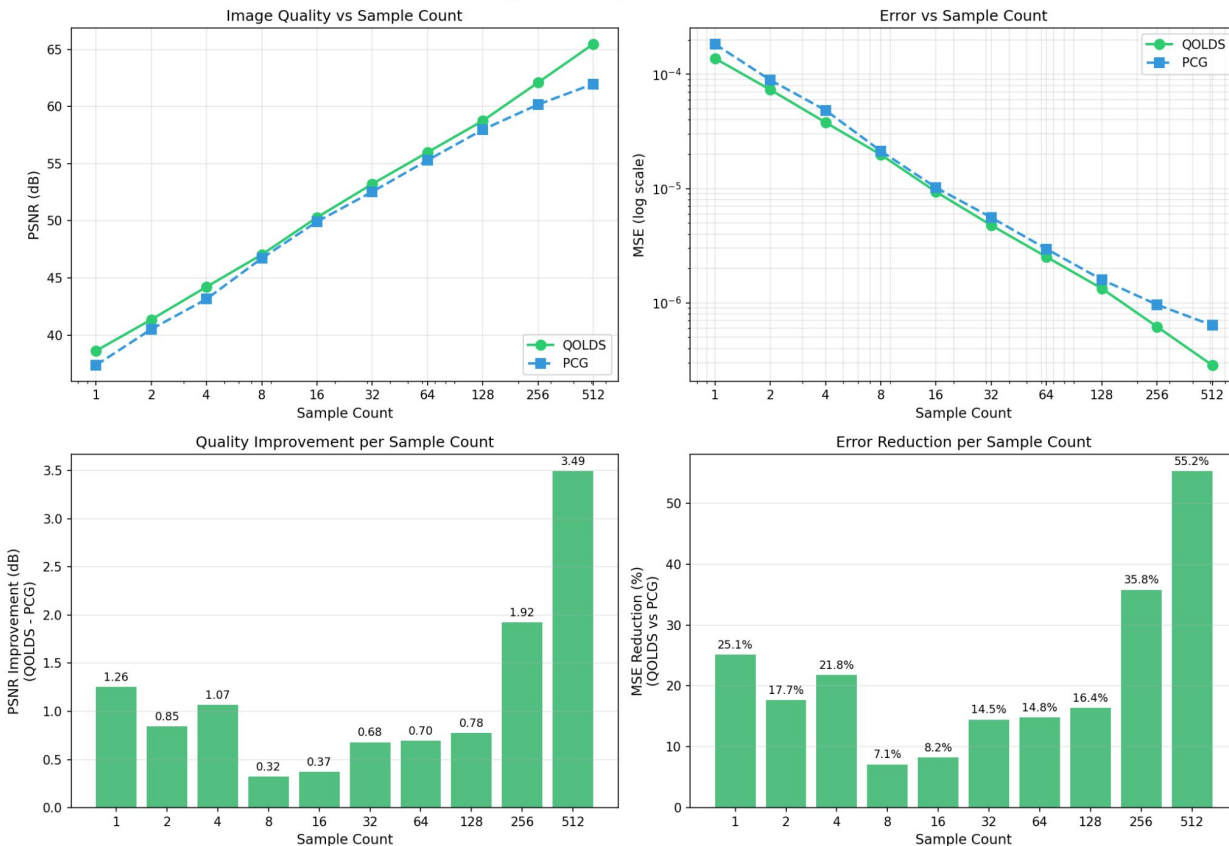
QOLDS - Overview



QOLDS - Comparison

QOLDS - Performance

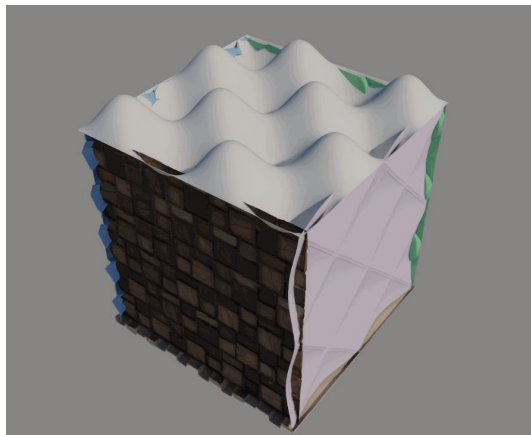
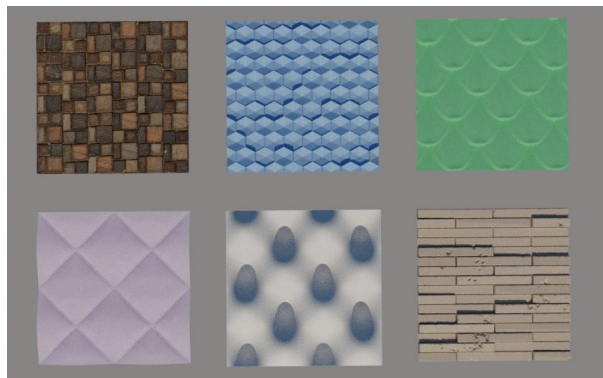
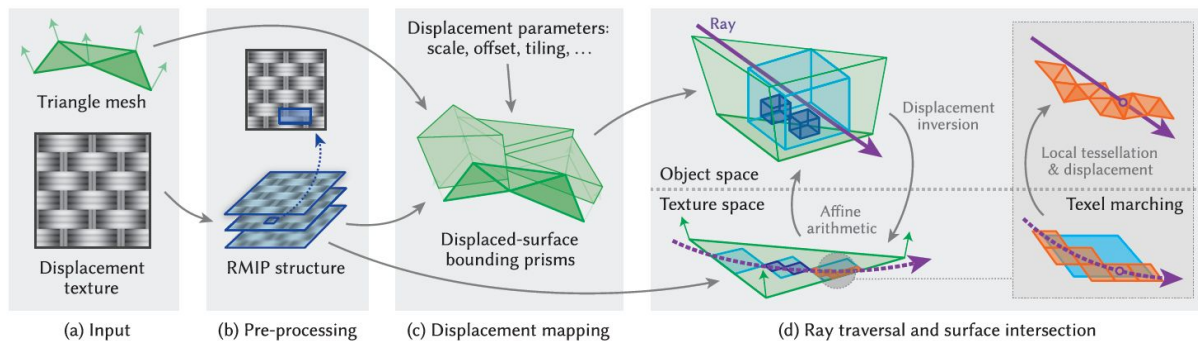
Convergence Analysis: QOLDS vs PCG



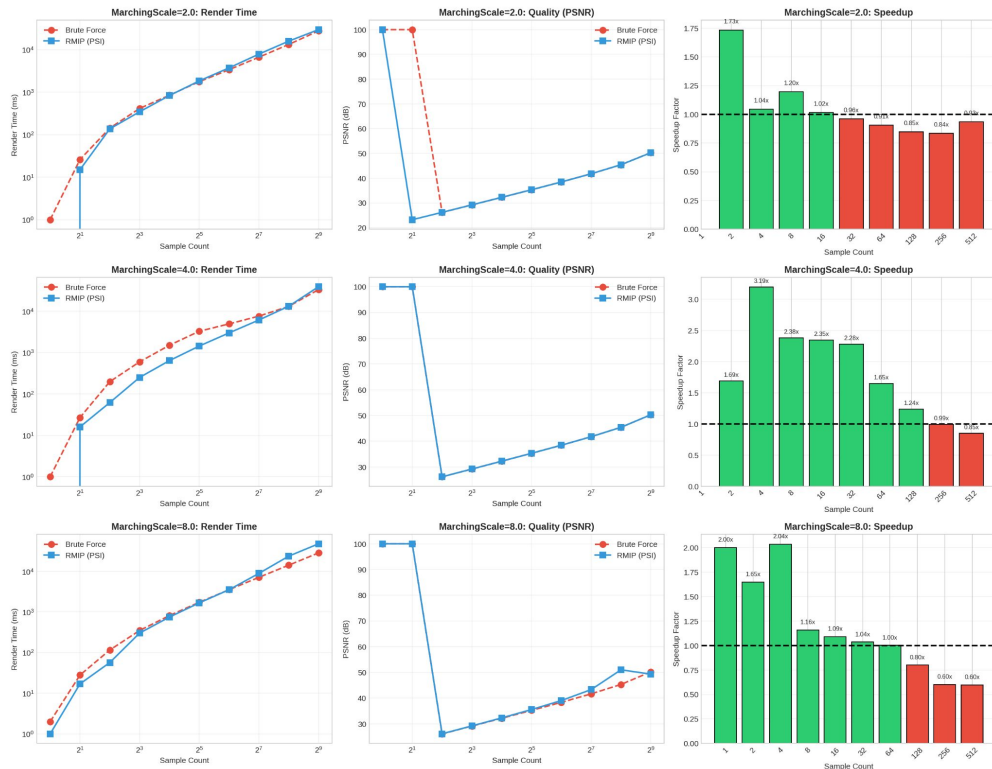
RMIP - "Rectangular MinMax Image Pyramid"

Tessellation-free displacement ray tracing

RMIP - Overview



RMIP - Performance



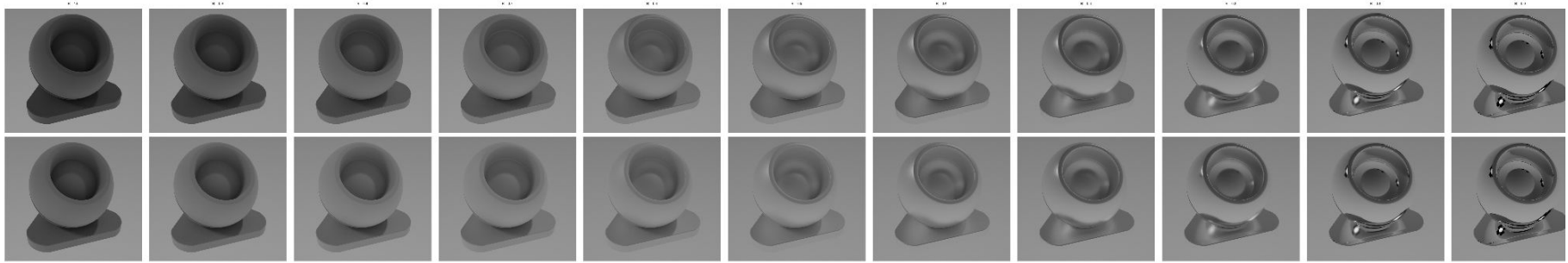
PSI-Marching vs Brute Force

- Low SPP:
PSI Marching provides 1.5x-4x speedup
- High SPP (128+):
PSI becomes slower due to overhead

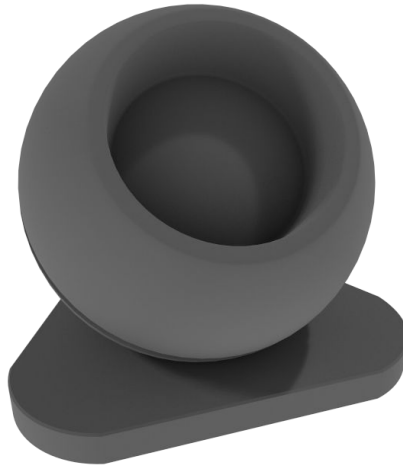
Fast-MSX: Fast Multiple Scattering Approximation

Compensate for the energy lost due to high roughness

Fast-MSX



GGX-Smith Only



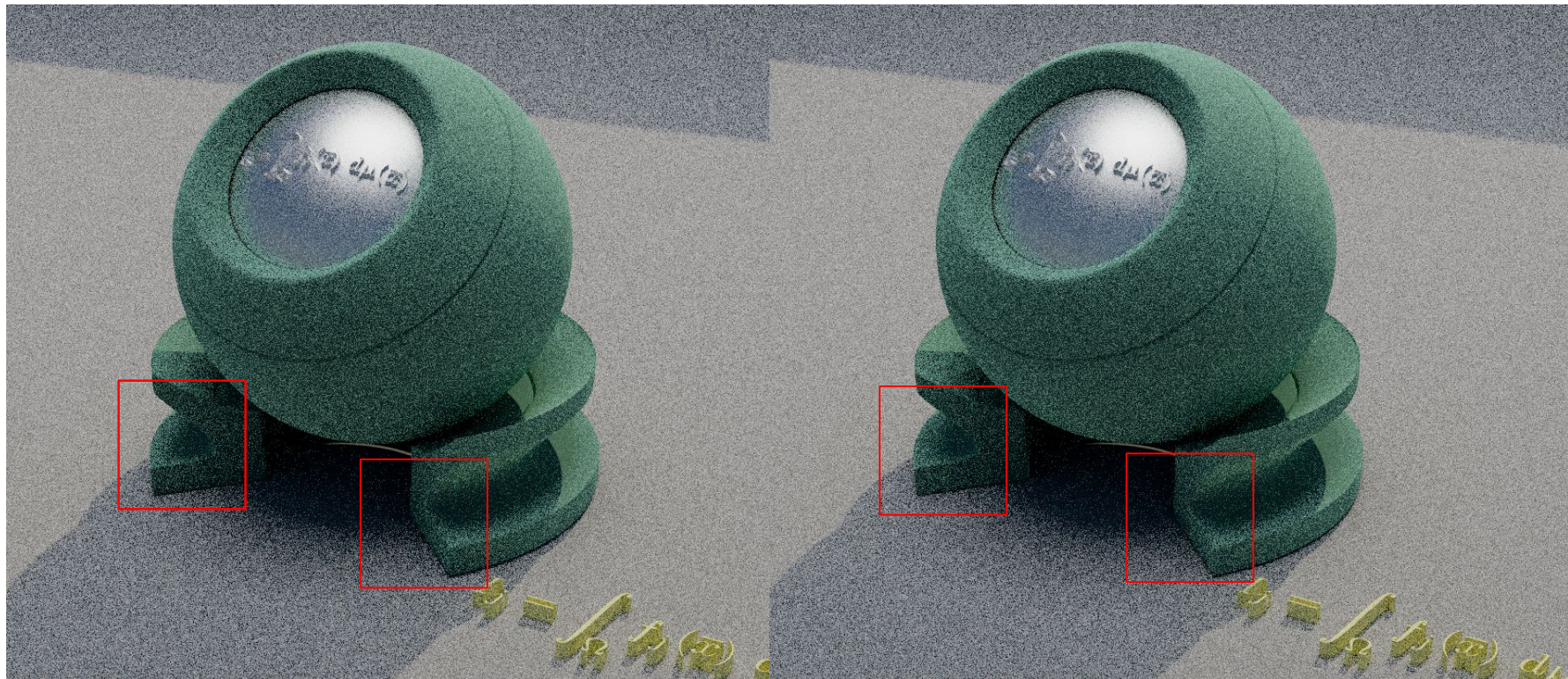
GGX-Smith & Fast-MSX, more bright

Roughness = 1

Bounded-VNDF

Clipping the sampling domain to exclude invalid, energy-loss paths

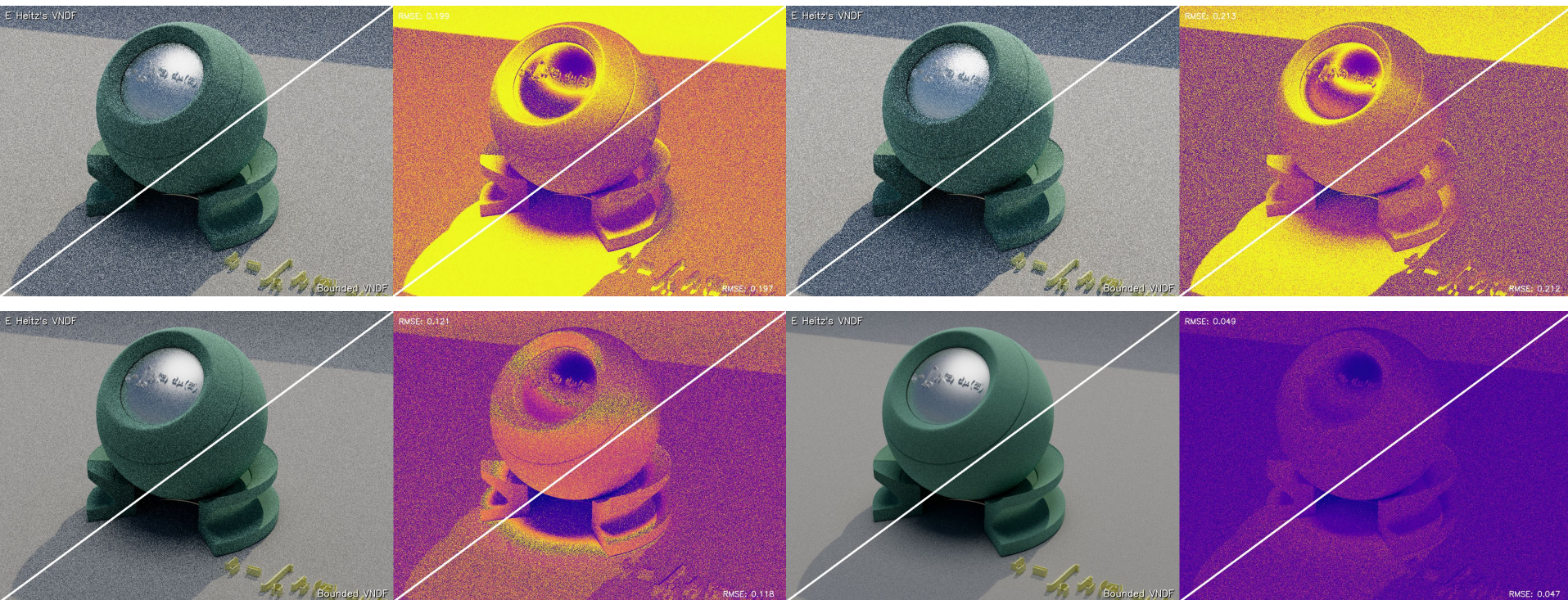
Bounded-VNDF



Standard VNDF

Bounded-VNDF, more details

Bounded-VNDF



Lower RMSE is achieved using Bounded-VNDF

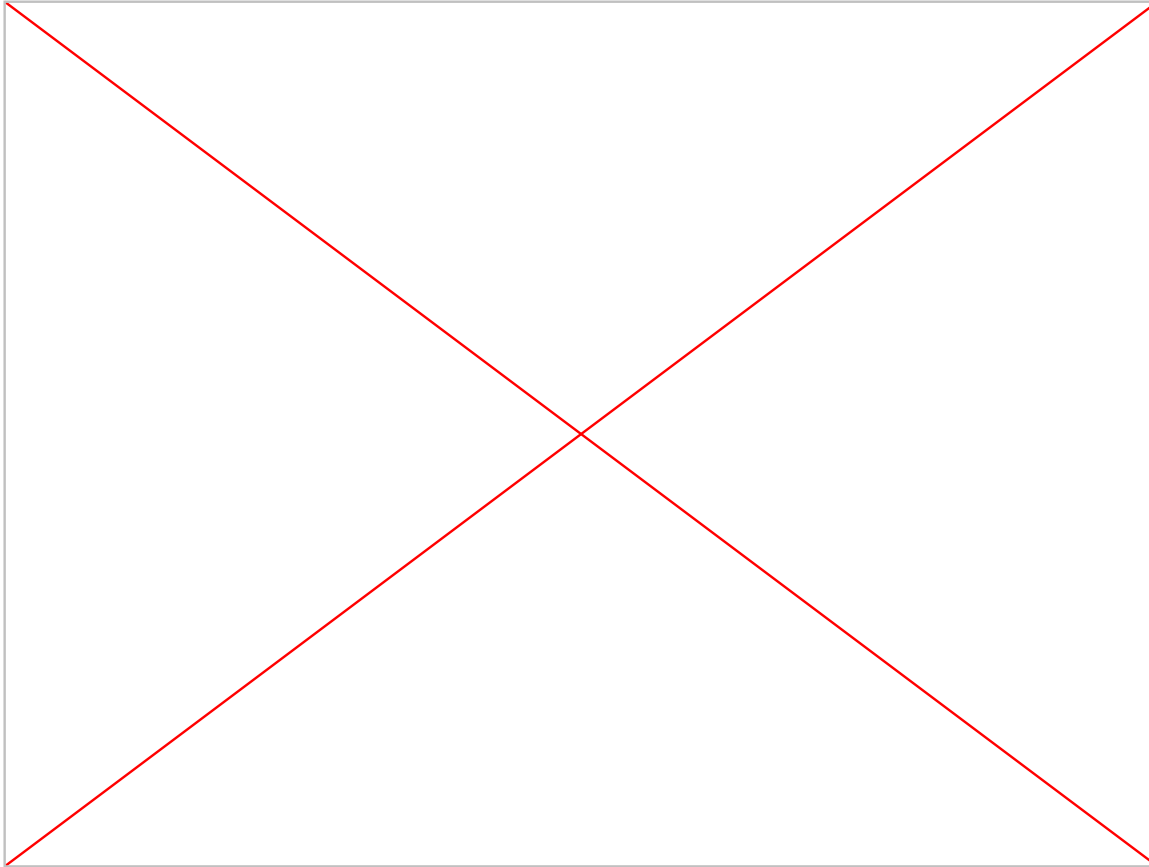
Final Demo Scene: The Forgotten Hall



Final Demo Scene: The Forgotten Hall



Demo Video



Thank You!

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